

Fruit Golf

An Asymmetrical Shared Space VR/Mobile Experience

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ABSTRACT

While the global COVID-19 pandemic did not catalyze widespread adoption of virtual reality (VR) technologies across all industries as some had anticipated, studies like Hall et al. from 2022 have demonstrated that public valuation of VR remains strongly in gaming, entertainment, and socializing [Hall et al., 2022].

As we look towards a future in which indoor gatherings with friends and family are safe and encouraged once again, there is an opportunity to position VR gaming as a go-to add-on to social gatherings by emphasizing ease of access for players of all levels of experience, and designing gameplay that encourages engagement rather than isolation in shared space.

Fruit Golf aims to use an asymmetric multiplayer format to offer an experience that spans collaborative and competitive experiences, and allows players to seamlessly interact with VR, mobile, and physical spaces in ways that most will have never seen before.

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1 INTRODUCTION

Graphically simulated golfing experiences have a long history, with indoor golf spaces popularizing in the late 1970s [Foreester and Beggs, 2001]. In 2000, Govin et al. made significant advancements with a video-based augmented reality golf simulator with head-mounted display [Govil et al., 2000]. Today, VR golf and mini golf simulators are commonplace across gaming platforms and provide both single player and multiplayer modes of play.

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Figure 1: Players competing against each other in Fruit Golf

Fruit Golf builds on this legacy while aiming to distinguish itself from a minigolf simulator, and instead fill a much-needed gap in shared space VR gaming. Fruit Golf disrupts the need for users to go out and seek entertainment venues with multiple headsets, and instead, offers the opportunity for groups to play fully with only one headset needed.

Much in the same way that Mario Kart has a broader appeal than racing simulations such as Forza or Gran Turismo, Fruit Golf takes established rules of a familiar sport, simplifies them, and places players in worlds with characters and interactions that they would never experience in real life.

2 DESIGNING FRUIT GOLF

While there exist other asymmetrical multiplayer experiences for virtual reality that use mobile devices, Fruit Golf is unique in that the players share the same coordinate space and are properly co-located together. In early user testing, we found that co-location in Fruit Golf contributes to a higher sense of social presence for both VR and mobile users.

Generally speaking, the asymmetric gameplay design of Fruit Golf is focused less on fairness, and more on motivating players to jump into the headset.

The VR player has ultimate control over the selection of the minigolf course. Courses are larger than life, with a clear commitment to different themes which inform the environment, music, interactable objects, characters, and obstacles in each level. When



Figure 2: Fruit Golf Concept Art from Pirate Ship Theme

play begins, players are introduced to different fruits, whose distinct physical characteristics contribute to varied behaviours when hit. The satisfaction of putting a fruit using natural motion with the controller is quickly heightened by the first obstacle placed in the VR players' path, and the pressure is on to make their way to the end of the course.

Simultaneously, mobile players need to collaborate quickly to prevent the VR player from attaining their goal. Mobile players have a bird's eye view on their VR opponent, and obstacles are listed as buttons on the right side of mobile player screens. Mobile players have access to only a certain number of obstacles at a time, and after a mobile player drops an obstacle, they need to wait for a few seconds before being able to access another. This means that placement is key, and communicating with their fellow mobile players is essential to maximize the challenge for the VR player.

Fruit Golf elevates the experience and joy of passively watching someone else experience and play VR, inviting casual audiences to bond together, strategize, and make real changes to the VR world using devices that they already have in their pocket.

3 TECHNICAL NOTES

Fruit Golf uses the built-in Unity physics features. A high degree of deterministic physics is used to synchronize multiplayer gameplay across multiple devices. To ensure a reliable golfing experience, the golf club is being iterated to react to any input from the user, including swing speed and hit accuracy.

Each fruit has specific values for the rigidbody like mass, drag, and angular drag, and has different physics materials, including friction and bounciness. Because each fruit will have unique characteristics, the club is designed so that it does not feel biased towards any one configuration.

The VR player is the server, and is authoritative. Mobile players are found using auto discovery on LAN using UDP broadcast. Mobile clients send a place obstacle command in its position/rotation relative to the VRs coordinate space. Networking for Fruit Golf is based on ENet. Each packet is its own class that handles its own serialisation so that in the future we can add compression to specific packets or even swap out the serialiser.

4 SETUP AND INSTALLATION

Fruit Golf requires approximately a 10x10 foot space. The Fruit Golf VR app is hosted on App Lab and can be played on the Oculus Quest

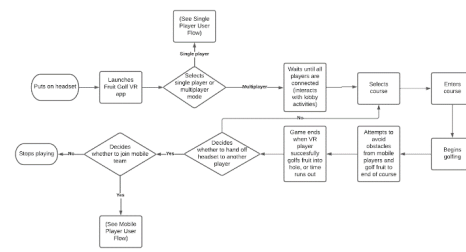


Figure 3: Multiplayer VR User Flow

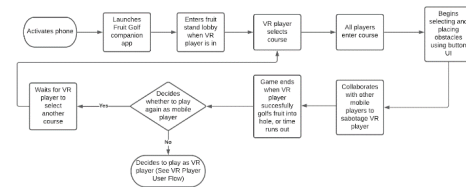


Figure 4: Multiplayer Mobile User Flow

and Vive Focus headsets. The Fruit Golf companion app is downloadable for iOS and Android. As soon as the VR app is launched, mobile players can connect and start engaging immediately. During exhibits, a large TV or projector is setup for spectators to view what the VR player is experiencing.

5 DISCUSSION

Using Fruit Golf as a testbed for creating social experiences through asymmetrical interactions between virtual reality users and mobile phone users allows us to determine which elements are most effective and generalize those solutions to areas outside of gaming, such as education and training.

Giving teachers and instructors the ability to track student locations within virtual environments and seamlessly control the placement of obstacles or other challenges allows VR technology to enhance existing human expertise, and has great potential in areas such as professional athletics, safety training, rehabilitation, and much more.

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